Chemistry 3A – Fall 2026 **Version A**

Sections 25810-25811

Midterm Examination #3 3 November 2026

Select **BEST** response for question. M/C questions are 2 pt except for calculation (quant) questions which are shown in brackets; to get full quant points you must show work on SHOW WORK page. All T/F are 1 pt. Be careful of precision (significant digits and decimal places).

1. Which of these statements IS a principle or assumption of Kinetic Molecular Theory of Gases?
 a) a gas is composed of tiny particles that have strong attractive forces
 b) gas particles collide elastically: not losing energy in collisions with each other and container walls
 c) gas particles are separated by distances very small relative to their size
 d) the average speed is dependent on the volume of the container

2. Potassium chloride (KCl) is put in a beaker of water. Which of these is a ***true*** statement about this solution?
 a) Water (H_2O) is the ***solute*** and KCl would be the ***solvent***
 b) In ***saturating*** aqueous solutions, KCl dissolves ***completely*** to form K^+ and Cl^- ions
 c) ***Polar*** **water** (H_2O) is better than ***nonpolar*** **hexane** (C_6H_6) in choosing to dissolve KCl
 d) KCl forms a ***precipitate*** when dissolved in water

3. Which of the statements below is a ***true*** statement regarding *electrolytes* and *non-electrolytes*?
 a) Compounds which are non-electrolytes are good conductors of electric current
 b) Glucose and ethanol are well-recognized as producing electrolytic solutions
 c) Compounds which are electrolytes form ions in solution
 d) Non-electrolytes are what makes batteries produce a voltage to create electricity

4. Convert a pressure (*P*) of 380 mm Hg to atmospheres (atm): **[3]**
 a) 0.50 atm b) 1.0 atm c) 380 atm d) 760 atm

5. Which mathematical relationship expresses the ideal gas law equation?
 a) $P_1V_1 = P_2V_2$ b) $P_1/T_1 = P_2/T_2$ c) $V_1/T_1 = V_2/T_2$ d) $PV = nRT$

6. Which of the statements is **TRUE** about **solutions** as discussed in lecture?
 a) *Air* is an example of a **heterogenous mixture**
 b) The **solute** is the *major component* of the solution.
 c) The ***solute*** in a solution of sodium chloride (NaCl) dissolved in water is NaCl
 d) A solution has ***two*** components: the *homogeneous* and the *heterogeneous* compounds

7. Which of the following is **TRUE** about pressure as related to gases?
 a) It is a factor in Gay-Lussac's Law
 b) It is understood in physics as a force per unit area (P=force/area)
 c) A measured pressure can be expressed in units of **torr**
 d) all of the above are true

8. A gas has an initial pressure of 1.00 atm and temperature of 25.0°C. The pressure is increased to 2.00 atm. What happened to the temperature if all other properties remain constant? **[4]**
 a) 12.5°C b) 50.0°C c) 298°C d) 323°C

9. A dilution of a C_1 = 1.00 *M* KCl stock solution was used to make a V_2 = 500.0 mL of a C_2 = 0.100 M KCl solution. What stock volume aliquot V_1 was used? **[3]**
 a) 5.00 mL b) 50 mL c) 500. mL d) 50.0 mL

10. Which of the following is **TRUE** about written chemical reactions?
  a) Conservation of mass and charge is required in the balancing of chemical reactions
  b) The chemical compounds to the left of the arrow (yield symbol) are called "products"
  c) A polyatomic molecule should ALWAYS be shown in parentheses no matter how many of these polyatomic molecules are present in a compound
  d) The *number* of the elements in any compound on both sides of the reaction arrow (⟶) can vary (be unequal)

11. Which of these reactions of barium chloride ($BaCl_2$) and sodium sulfate (Na_2SO_4) shows the ***net ionic equation***?
  a) $BaCl_2\,(aq) + Na_2SO_4\,(aq) \longrightarrow BaSO_4\,(s) + 2\,NaCl\,(aq)$
  b) $Ba^{2+}\,(aq) + 2\,Cl^-\,(aq) + 2\,Na^+\,(aq) + SO_4^{2-}\,(aq) \longrightarrow BaSO_4\,(s) + 2\,Na^+\,(aq) + 2\,Cl^-\,(aq)$
  c) $Ba^{2+}\,(aq) + SO_4^{2-}\,(aq) \longrightarrow BaSO_4\,(s)$
  d) NONE of the above show a net ionic equation

12. What does this reaction show: $Zn \longrightarrow Zn^{2+} + 2e^-$?
  a) This is a *complete chemical equation*
  b) This is a *complete ionic equation*
  c) This reaction is a *net ionic equation*
  d) This is a *half-reaction*

13. An ideal gas with amount n = 1.00 mol, volume V = 10.0 L, and temperature T = 100.0°C. Calculate the pressure P in atmospheres (atm)? **[5]**
  a) 0.306 atm b) 3.06 atm c) 3.100 atm d) 30.6 atm

14. Precipitation reaction $AgNO_3\,(aq) + NaCl\,(aq) \longrightarrow AgCl\,(s) + NaNO_3\,(aq)$ is shown. Which of the statements below is **FALSE** about the process of balancing this reaction?
  a) Balancing can be done by changing subscripts of compounds, like $AgNO_3$ to $Ag(NO_3)_2$
  b) The reaction equation should only ever be balanced by adjusting the whole-number coefficients of the compounds
  c) Adding or removing any compound on either side of the reaction arrow must never be done to achieve balancing.
  d) Changing the phases of the compounds, such as changing from aqueous (*aq*) to solid (*s*) or to gas (g) has nothing to with balancing.

15. Which reaction type is this reaction? $Mg\,(s) + Cu(NO_3)_2\,(aq) \longrightarrow Mg(NO_3)_2\,(aq) + Cu\,(s)$
  a) precipitation b) halogen replacement c) metal replacement d) combination

16. 10.0 g of sodium bromide (NaBr) [molar mass = 102.89 g/mol] is used to make a 100.0 mL in water. What is the concentration of the NaBr solution? **[4]**
  a) 0.0972 *M* b) 0.486 *M* c) 0.972 *M* d) 1.94 *M*

17. This is written on the whiteboard: "$Fe^{2+} \longrightarrow Fe^{3+} + e^-$". Which of the statements is **TRUE**?
  a) This shows a reduction half-reaction
  b) This shows an oxidation half-reaction because the reactant Fe^{2+} is **gaining** an electron
  c) Aside from the reaction shown, every oxidation half-reaction must have an accompanying or associated reduction half-reaction
  d) The arrow as shown is never used to show this type of reaction

18. Which of these compounds as solutes will have a *greater* effect on lowering/depressing the freezing point temperature of a solvent given they completely dissolve?
a) KCl b) $CaCl_2$ c) ethanol (CH_3CH_2OH) d) glucose ($C_6H_{12}O_6$)

19. 2.00 mol of zinc chloride ($ZnCl_2$) [molar mass = 136.29 g/mol] is dissolved in final 0.500 kg water. What is the molality of the $ZnCl_2$ solution? **[4]**
a) 0.500 *m* b) 1.00 *m* c) 2.00 *m* d) 4.00 *m*

20. Which reaction below shows a **halogen replacement** (*single replacement*) reaction type?
a) Cl_2 (*g*) + 2 NaBr (*aq*) ⟶ 2 NaCl (*aq*) + Br_2 (*l*)
b) Mg (*s*) + $Cu(NO_3)_2$ (*aq*) ⟶ $Mg(NO_3)_2$ (*aq*) + Cu (*s*)
c) $BaCl_2$ (*aq*) + Na_2SO_4 (aq) ⟶ $BaSO_4$ (*s*) + NaCl (*aq*)
d) H_2O (*l*) ⟶ H_2 (*g*) + 2 O_2 (*g*)

21. Which statement is **TRUE** with respect to colligative properties of a solution?
a) **NaCl** and **$MgCl_2$** will have different effects because properties of the compound are important
b) Knowing the **molarity** (*M*) is required in calculating freezing and boiling point changes.
c) A solution's change in boiling point and freezing point are examples of colligative properties
d) A compound that produces three ions in solution and another producing two ions will have the same effect on the boiling point

22. Find the **FALSE** statement concerning the oxidation number of molecule nitrogen dioxide (NO_2)?
a) The oxidation number of oxygen (O) atoms is -2 (minus 2), which is typical
b) The sum of all oxidation numbers for NO_2 is 7
c) The oxidation number of the nitrogen (N) atom is +5
d) NO is a molecule in which its nitrogen (N) will have a different oxidation number than N in NO_2

23. Copper (Cu) metal only with the strongest of acids like nitric acid HNO_3 (which generate H^+) to produce copper ion (Cu^{2+}) and hydrogen gas (H_2). Which of the reactions below is correctly balanced and shows what occurs?
a) 2 Cu (*s*) + 2 HNO_3 (*aq*) ⟶ H_2 (*g*) + $CuNO_3$ (*aq*)
b) 2 Cu (*s*) + 2 HNO_3 (*aq*) ⟶ H_2 (*g*) + $Cu(NO_3)_2$ (*aq*)
c) Cu (*s*) + 4 HNO_3 (*aq*) ⟶ H_2 (*g*) + $Cu(NO_3)_2$ (*aq*) + 2 NO_2 (g)
d) Cu (*s*) + HNO_3 (*aq*) ⟶ O_2 (*g*) + $Cu(NO_3)_2$ (*aq*)

24. The propane (C_3H_8) in a gas grill reacts with oxygen (O_2): what type of reaction is this?
a) combustion
b) acid-base neutralization
c) combination
d) precipitation

25. Which one of these reaction types shows a double replacement reaction?
a) C_3H_8 (*g*) + O_2 (*g*)⟶ CO_2 (*g*) + H_2O (*g*)
b) $FeCl_2$ (*aq*) + Ca (s) ⟶ CaF_2 (*aq*) + Fe (*s*)
c) $CaCO_3$ (*aq*) ⟶ CaO (*aq*) + CO_2 (*g*)
d) 2 KI (*aq*) + $Pb(NO_3)_2$ (*aq*) ⟶ PbI_2 (*s*) + 2 KNO_3 (*aq*)

CONTINUE TO TRUE-FALSE QUESTIONS ON NEXT PAGE

TRUE or FALSE questions. (a) for true, (b) for false. 1 point each

26. **HCl (*aq*) + NaOH (*aq*)** is a classic acid-base neutralization reaction.
a) true b) false

27. The *oxidation number* of *oxygen* atom in most cases is 2– (or -2) based on its electronegativity.
a) true b) false

28. To calculate the **volume percent** of a solution, the solute should be a *liquid*.
a) true b) false

29. The **molarity** of a solution is *number of moles* divided by the *volume in liters*.
a) true b) false

30. The formation of a precipitate is a reaction type that means a gaseous compound is formed as a product.
a) true b) false

31. Boyle's Law relates two properties of gases: pressure and volume.
a) true b) false

32. For the reaction Zn (*s*) + 2 HCl (*aq*) → H_2 (*g*) + $ZnCl_2$ (*aq*), H_2 is a *reactant*.
a) true b) false

33. Ionic compounds dissolve in polar solvents and polar compounds do not dissolve in nonpolar solvents
a) true b) false

34. When an element ***loses*** electrons in a reaction, this is called an ***oxidation***.
a) true b) false

35. Osmosis describes a process when salts move through membranes to make water volumes equal on both sides of the membrane
a) true b) false

The SHOW YOUR WORK page is next sheet. Use both sides. If you need more sheets, raise your hand to request more

Quantitative M/C SHOW-YOUR-WORK page. **Name**_____ **ID** _____
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused